Digital Signal Processing

Fourier Series and Transforms

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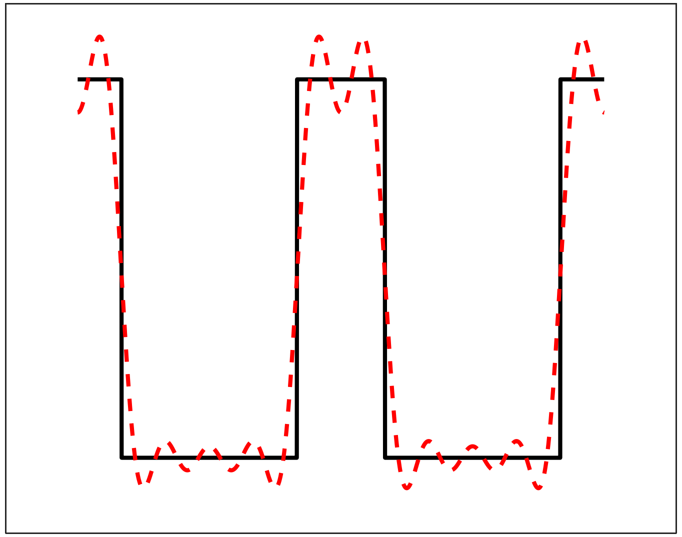
A signal can be viewed in time or frequency

The basis functions of *Fourier analysis tools* are sinusoids

Unlike the DFT, these sinusoids have infinite duration

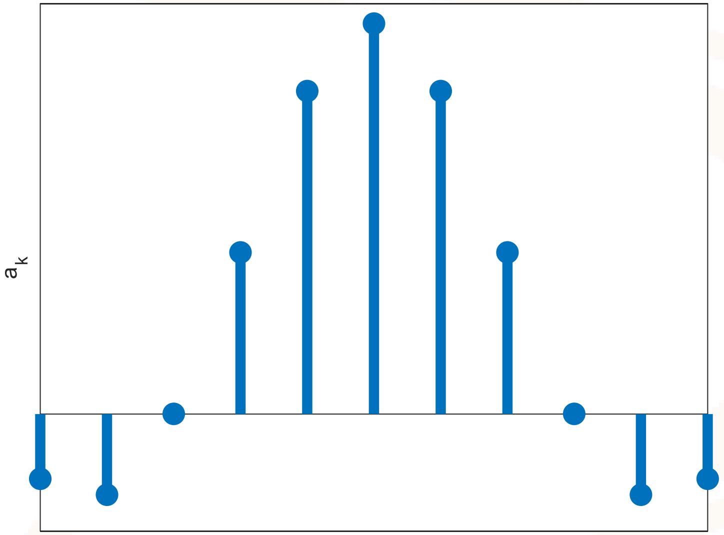
$$x(t) = x(t+T)$$

$$x(t) = \sum_k c_k\, exp(jk2\pi t/T)$$



A

A^{-1}



Fourier Series are "a special case" of Fourier Transforms

Best *tool* to the signal being analyzed

	Continuous-Time	Discrete-Time
Periodic	Fourier Series (FS) [V]	Discrete-Time Fourier Series (DTFS) [V]
Non-periodic	Fourier Transform (FT) [V/Hz]	Discrete-Time Fourier Transform (DTFT) [(2π) V/rad]

Duality between Time and Frequency domains

First Domain			Second Domain
Periodic			Discrete
Non-periodic			Continuous
Periodic/Continuous Signal (FS)	$\int$	Σ	Discrete/Non-periodic spectrum
Non-periodic/Discrete Signal (DTFT)	Σ	$\int$	Continuous/Periodic spectrum

Duality between Time and Frequency domains

The big picture

	Continuous Time	Discrete Time
Fourier Series	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$ continuous and periodic in time $a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$ discrete and aperiodic in frequency	$x[n] = \sum_{k=<N>} a_k e^{jk\frac{2\pi}{N}n}$ discrete and periodic in time $a_k = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk\frac{2\pi}{N}n}$ discrete and periodic in frequency
Fourier Transform	$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$ continuous and aperiodic in time $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$ continuous and aperiodic in frequency	$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$ discrete and aperiodic in time $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$ continuous and periodic in frequency

Table 4.1: Comparison of Fourier series and Fourier transform.

https://drive.google.com/file/d/1ov_X0a7tlJzuVRJS1_mJLblwnpACoeXK/view

This duality is not entirely consistent. In block transforms, the DFT spectrum is always periodic, and it is still referred to as a "transform".

Fourier Series (FS)

Basis function
$e^{jk\omega_0 t}$

Infinite complex <u>harmonic</u> sinusoids

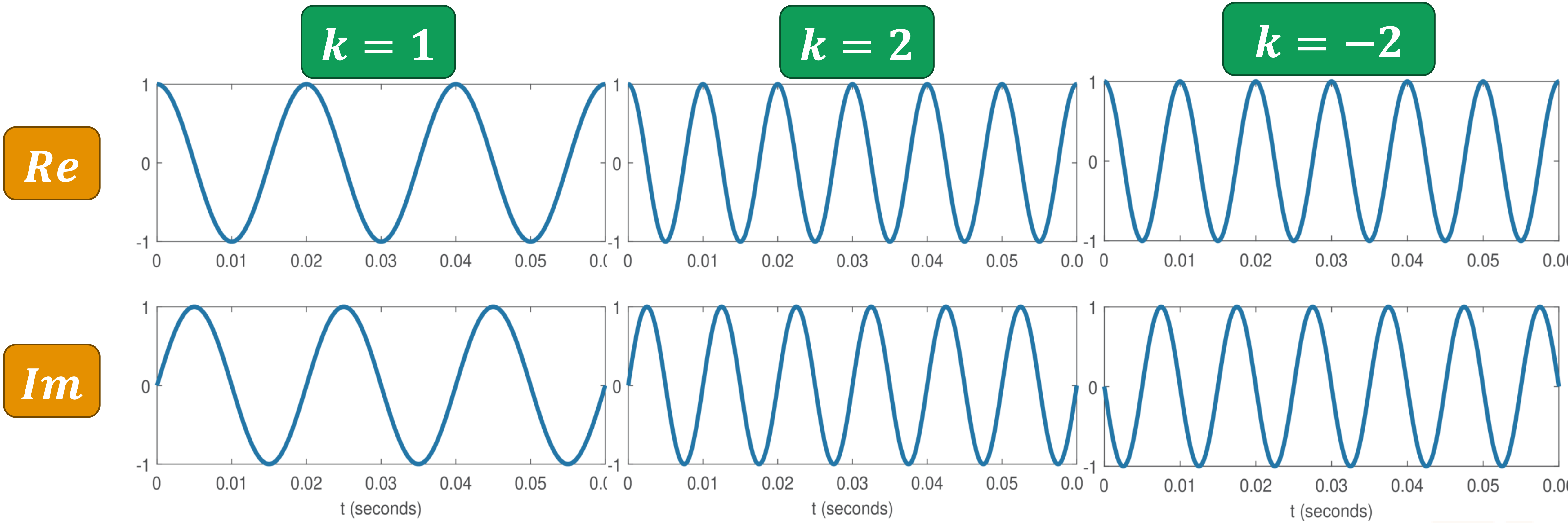
$$c_k = \frac{1}{T_0}\int_{\langle T_0\rangle} x(t)e^{-jk\omega_0 t}\,dt$$

$$c_k \Rightarrow \omega = k\omega_0 = k2\pi f_0$$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

$k = 0 \Rightarrow$ DC level
$k = \pm 1 \Rightarrow$ Fundamental Frequency
$k = \pm 2, \pm 3 \ldots \Rightarrow$ Harmonics

Basis functions of FS



Notice $k = \pm 2$ has same real part but opposite signs at imaginary part, due to even/odd symmetry of sines and cosines

Analyzing a real signal then yields Hermitian symmetry:

$$c_{-k} = c_k^*$$

One-Look Fourier Series (FS)

A sum of sinusoids yields coefficients by inspection

$$\cos(k\omega_0 t) = \frac{e^{jk\omega_0 t} + e^{-jk\omega_0 t}}{2}$$

$$\Rightarrow c_k = c_{-k} = 1/2$$

$$\cos(k\omega_0 t + \phi)$$
$$c_k = 0.5e^{j\phi}$$
$$c_{-k} = 0.5e^{-j\phi}$$

$$x(t) = 3 + \cos(3t) + 7\sin(5t)$$

$\omega_0 = 1$ rad/s

$c_0 = 3$

$c_3 = 0.5$

$c_{-3} = 0.5$

$$c_5 = 3.5e^{-j\pi/2} = -3.5j$$

$$c_{-5} = 3.5e^{j\pi/2} = 3.5j$$

Other $c_k = 0$

Real signals can also be represented with unilateral spectrum, but the bilateral spectrum is emphasized in the text

Discrete-time Fourier Series (DTFS)

Basis function
$e^{jk\Omega_0 n}$
$\Omega_0 = 2\pi/N_0$ rad

N_0 **distinct complex harmonic sinusoids**

Periodic spectrum
$X[k] = X[k+N_0]$

$$X[k] = \frac{1}{N_0}\sum_{n=\langle N_0\rangle} x[n]e^{-jk\Omega_0 t}$$

$$x[n] = \sum_{n=\langle N_0\rangle} X[k]e^{jk\Omega_0 t}$$

$$x[n] = A\cos\left(\frac{\pi}{5}n + \frac{\pi}{3}\right)$$

$$\Omega_0 = \frac{2\pi}{10},\ N_0 = 10$$

$$X[1] = \frac{A}{2}e^{\frac{j\pi}{3}}; X[-1] = \frac{A}{2}e^{-\frac{j\pi}{3}}$$

$$X[0] = X[2] = X[3] = X[4] = X[5] = X[6] = X[7] = X[8] = 0$$

$$X[k] = X[k+10]$$

DFT/FFT to DTFS

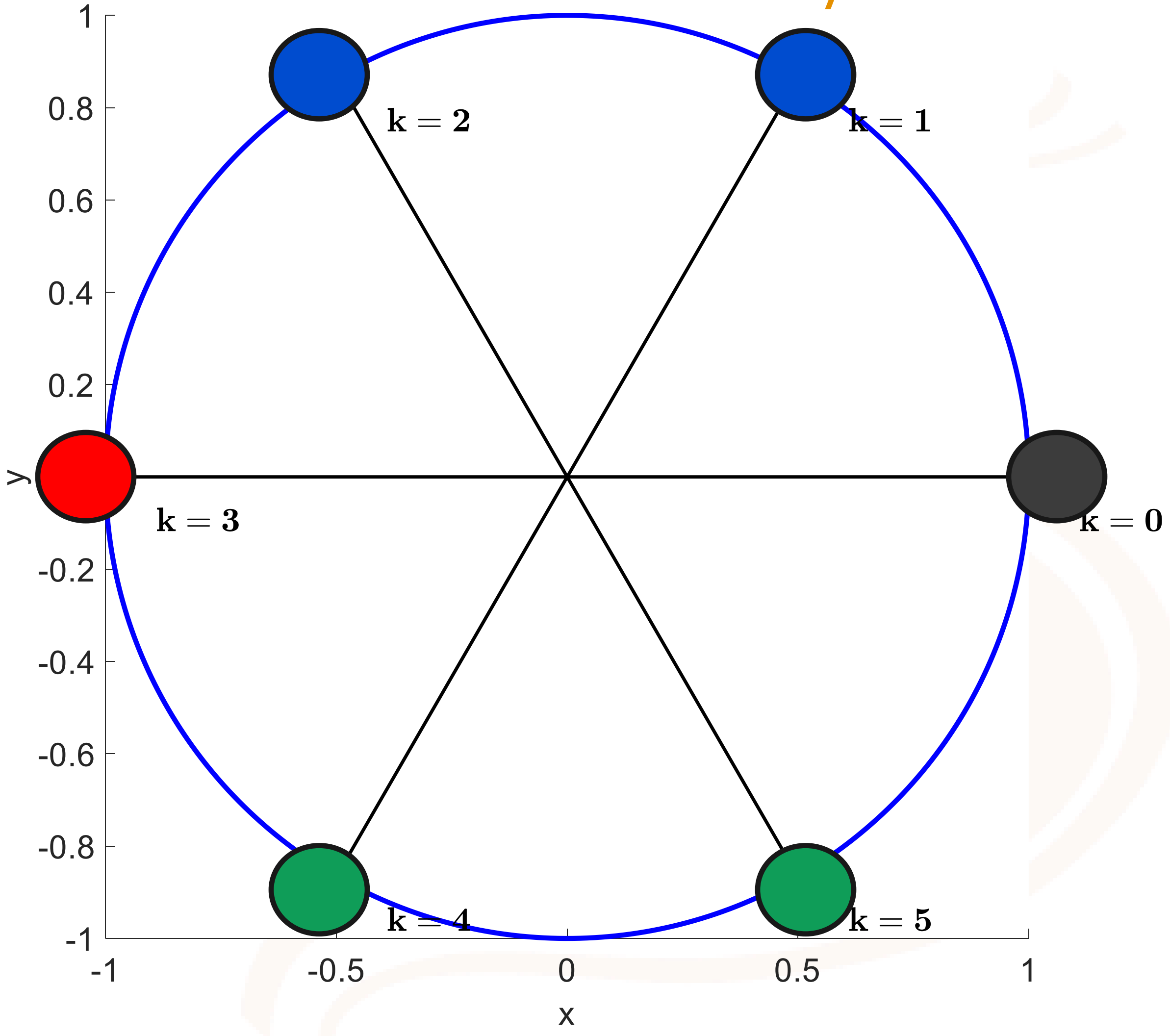
When $N = kN_0$

$$DTFS\{x[n]\} = \frac{DFT\{x[n]\}}{N}$$

k =	0	1	2	3	4	5
$X[k] =$	0	-1.2j	0.6	0	0.6	1.2j
$DTFS =$	0	-0.2j	0.1	0	0.1	0.2j

$$x[n] = 0.4sin\left(\frac{2\pi n}{6}\right) + 0.2cos\left(\frac{4\pi n}{6}\right)$$

$\Delta\Omega = \pi/3$



Fourier Transform (FT)

Basis function
$e^{j2\pi f t}$

Continuum frequency (V/Hz)

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} \mathrm{d}t$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} \mathrm{d}f$$

Linear Frequency

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \mathrm{d}t$$

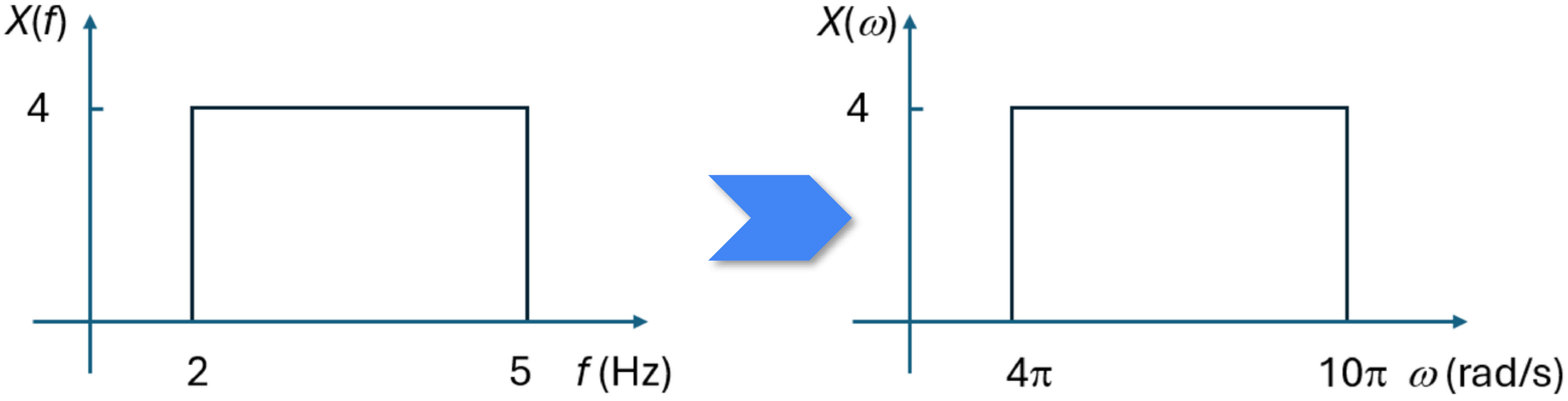
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} \mathrm{d}\omega$$

Angular Frequency

Fourier Transform (FT) – Linear to Angular Frequencies

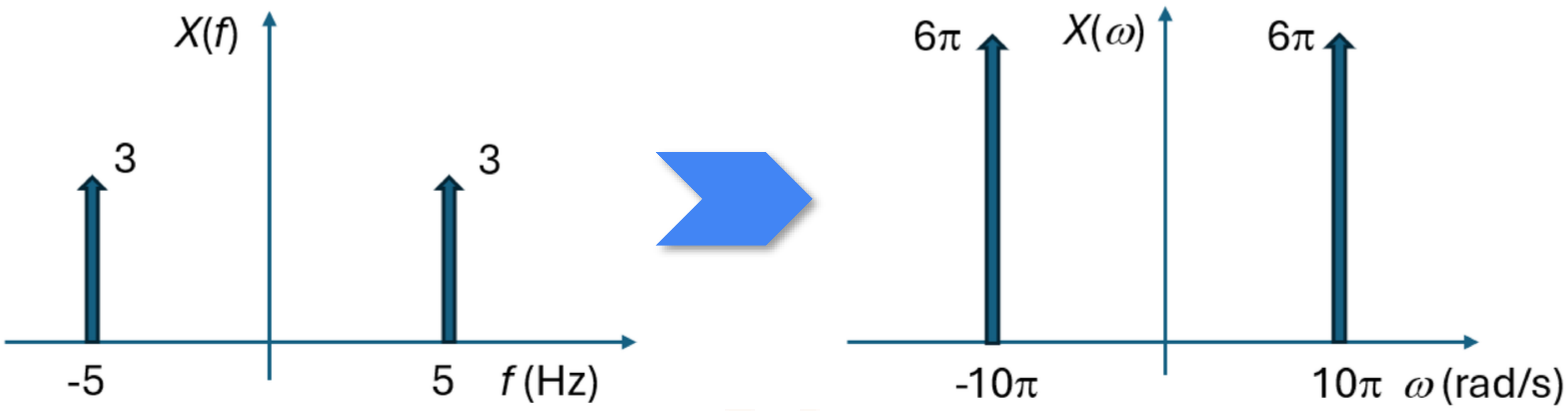
Without impulses

$$X(\omega) = X\left(\frac{f}{2\pi}\right)$$

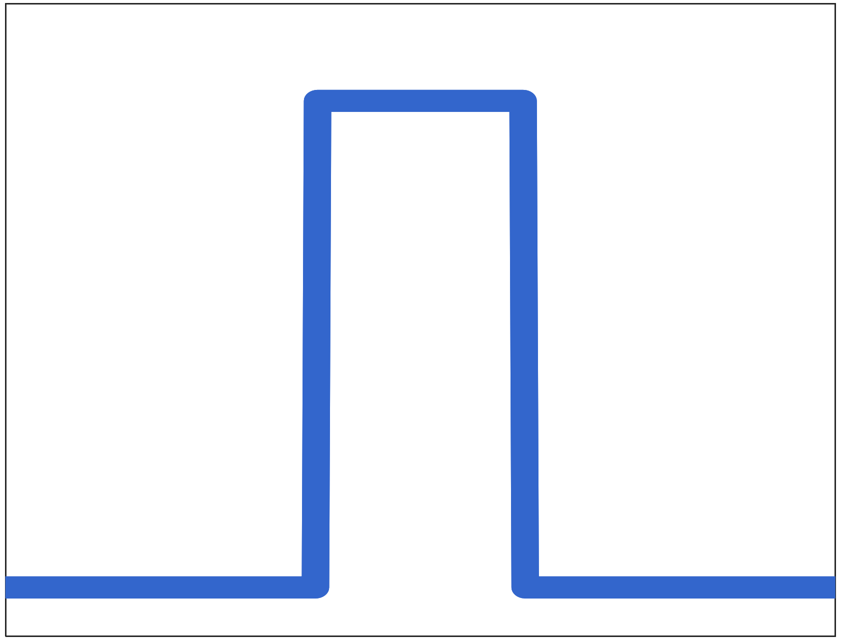


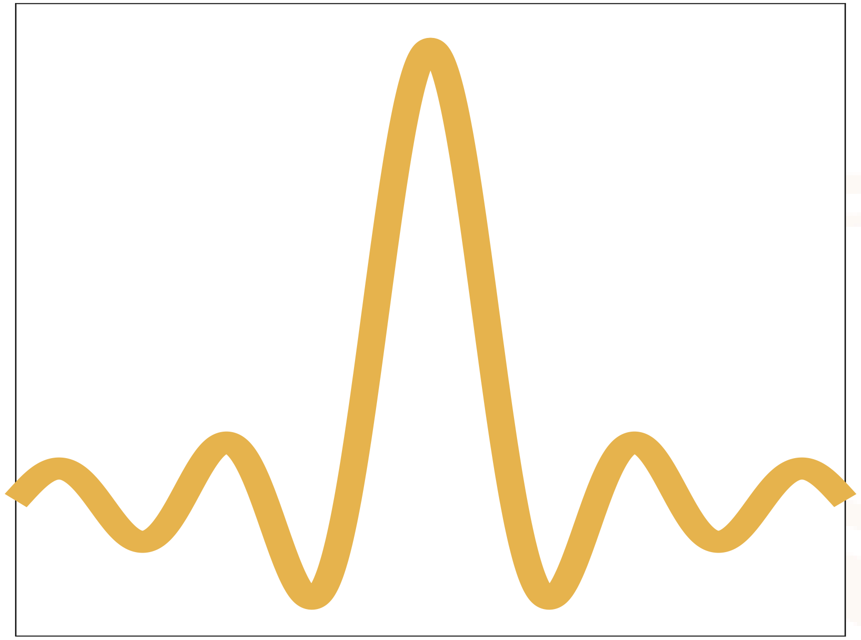
With impulses

Also scale impulses by 2π



Finite in one domain means infinite in another





rect $\xleftrightarrow{\text{FT or IFT}}$ sinc

finite $\xrightarrow{\text{FT or IFT}}$ infinite

Discrete-time Fourier Transform (DTFT)

Basis function
$e^{j\Omega n}$

Continuous and periodic frequency
(2π) V/rad

$$X(e^{j\Omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

Some Properties

$x[n]$	$X(\Omega)$
$x[n-n_0]$	$X(\Omega)e^{-j\Omega n_0}$
$\alpha\delta[n]$	α

DFT/FFT to DTFT

DFT/FFT is a sampling of DTFT

$$X[k] = X(e^{j\Omega})\Big|_{\Omega=\frac{2\pi}{N}k}$$

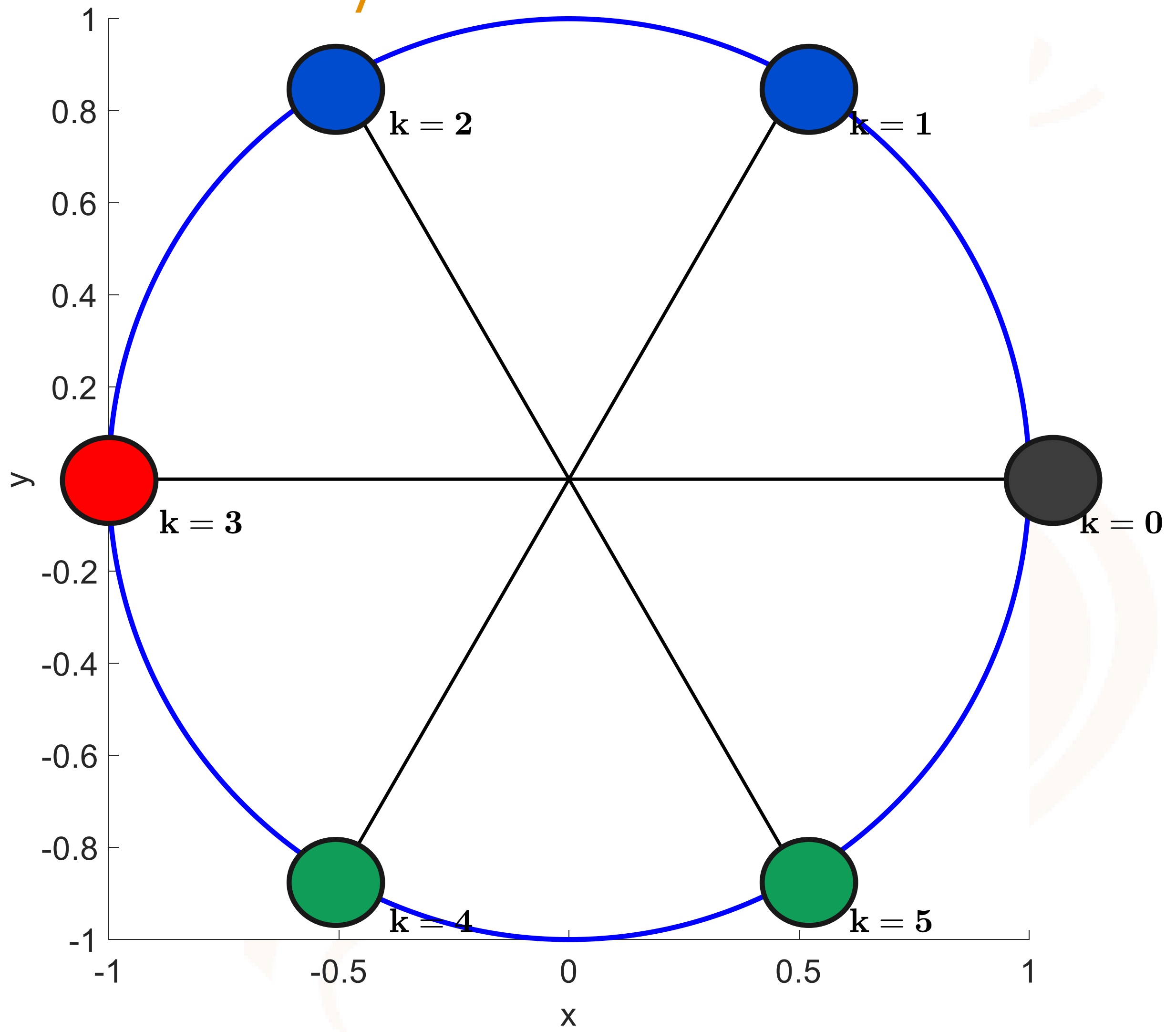
$$\Delta\Omega = 2\pi/N$$

$\{0, \Delta\Omega, 2\Delta\Omega, \ldots, (N-1)\Delta\Omega\}$ **are called DFT bins**

DTFT→ a continuous curve on the unit circle
DFT/FFT → "taking snapshots" of that curve

$\Delta\Omega = \pi/3$

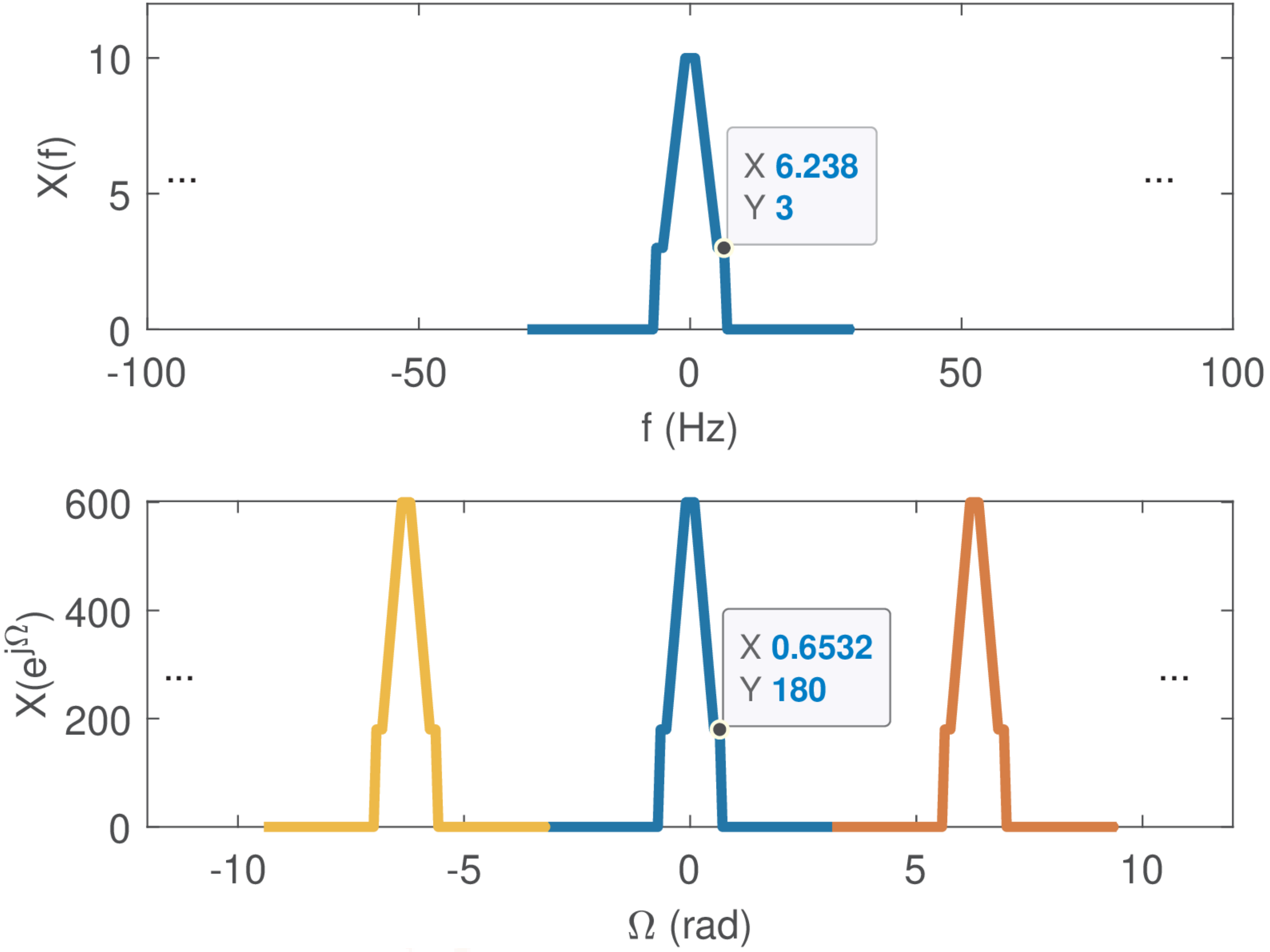
6 bins



AD process generate copies of the original spectrum every 2π

When sampling theorem is obeyed, there is no *aliasing*:

$$X\left(e^{j\Omega}\right) = F_s X(\omega)\Big|_{\omega=F_s\Omega}$$



Thank you!

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